

A Higher Kick Frequency Swimming Training Program Optimizes Swim-to-Cycle Transition in Triathlon

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Abstract

Ambrosini, L., Presta, V., Vitale, M., Menegatti, E., Guarnieri, A., Bianchi, V., De Munari, I., Condello, G., and Gobbi, G. A higher kick frequency swimming training program optimizes swim-to-cycle transition in triathlon. *J Strength Cond Res* XX(X): 000–000, 2023—The purpose of this study was to evaluate the effect of an 8-week swimming training program on biomechanical and physiological responses during a swim-to-cycle simulation. Fifteen triathletes were randomly allocated to 3 groups: a 6-beat-kick group (K6), a 4-beat-kick group (K4), and a control group (CG). Biomechanical and physiological parameters were evaluated during a 400-m swim and a 10-minute cycle segment before (Pretraining) and after (Posttraining) the program. A lower stroke frequency ($p = 0.004$) and a higher stroke length ($p = 0.002$) was found in K6 compared with CG at Posttraining. A reduction in the K6 emerged between Pretraining and Posttraining during cycling for heart rate ($p = 0.005$), $\dot{V}O_2$ ($p = 0.014$), and energy expenditure ($p = 0.008$). A positive association emerged between swim kick index and cycling cadence in the K6 group. The improvement in stroke frequency and length observed in the K6 group could be explained as an improvement in swimming technique. Similarly, the reduction in energy expenditure during cycling at Posttraining for the K6 group suggests an improvement in the working economy. Triathlon coaches and athletes should consider the inclusion of high swim kick into their training programs to enhance swim and cycling performance, which can ultimately lead to an improvement in the swim-to-cycle transition and the overall triathlon performance.

Key Words: swim kick, swim stroke, stroke frequency, kick frequency, cycling cadence, cycling power

Introduction

Triathlon competition is a combination of 3 disciplines (swimming, cycling, and running) linked by 2 transition periods. Several races are currently organized, including short-distance triathlon (i.e., Mixed-Team Relay, Sprint Distance, and Olympic Distance) and long-distance triathlon (i.e., Ironman 70.3 and Full Ironman).

The multidisciplinary nature of the triathlon performance induces a unique profile with biomechanical and physiological changes occurring during the swim-to-cycle (T1) (1) and cycle-to-run (T2) (26) transition, with the influence of a previous segment on the subsequent segment based on the applied pacing strategy. Considering the swim-to-cycle transition for the short-distance triathlon, it has been demonstrated that greater cycling power is required during the first meters of the second segment for slower swimmers with the attempt to bridge the gap to the front group (31). Differently, a greater intensity during the final part of the swim segment could be beneficial for a more efficient cadence during the first kilometers of the cycle segment (30), hence having an influence also on the subsequent run segment and on the entire triathlon performance (3).

Swimming performance is characterized by a coordinated action between the upper and lower limbs to produce a propulsive action, with the general assumption of the main contribution of the upper limbs for the body propulsion (16). However, recent studies emphasized the contribution of lower limbs causing a more effective action of the upper limbs, a lower inclination of the trunk, and a reduction of the passive resistance of the water, hence influencing the overall swimming performance (18,35). However, it is a common strategy among triathletes to use 2 swim kicks each stroke cycle, with the intention to save energy for the subsequent cycle segment, although a few of them adopt a 6-kick strategy. Therefore, it is hypothesized that an insufficient use of the legs during the swim segment could be detrimental for the following cycle segment (1,17). This hypothesis is based on the assumption that a previous high-intensity exercise (i.e., swim kicks) before a subsequent action (i.e., cycling) might elicit changes on physiological parameters, such as the improvement in muscle perfusion, the increase in residual acidemia, a better $\dot{V}O_2$ kinetics, and muscle fibers recruitment (6,17), and on biomechanical parameters, such as an increase in the cycling peak power and in the total work expressed during the first minutes of the cycling performance (8).

Focusing on physiological parameters, the main changes in $\dot{V}O_2$ kinetics are because of the muscle site involved (4). Therefore, a greater involvement of the muscles of legs during the exercise is expected to reduce the amplitude of the slow component

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Journal of Strength and Conditioning Research 00(00)/1–9

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of $\dot{V}O_2$ and increase the tolerability to the next exercise (21). Furthermore, the accumulation of metabolites (i.e., H^+ , lactate, K^+) because intense exercise performed with the arms (during the swim segment) could also impair muscles less involved, causing fatigue or affecting motor unit recruitment during the subsequent exercise performed with the legs (during cycle segment) (20). However, the chronic training effects on physiological and biomechanical parameters are poorly investigated. Furthermore, evidence is mainly available for the effect of the cycling cadence on running performance, although limited evidence is available for the effect of the swimming kicks on cycling performance.

Therefore, the purpose of this study was to evaluate the chronic effects of the swim kick strategy on physiological parameters (oxygen consumption, heart rate [HR], energy expenditure, and blood lactate) and biomechanical parameters (swim stroke and kick frequency, swim stroke and kick length, swim stroke and kick index, cycling cadence and power) through a specific 8-week training program in high-level triathletes. Furthermore, a secondary aim was to verify the existence of associations between swimming and cycling performance in the investigated biomechanical and physiological variables. It was hypothesized that a specific training program centered on a greater involvement of legs during the swim segment could positively affect the subsequent cycle segment because of a greater lower-body muscle activation and recruitment that leads to a better muscle perfusion and a reduction in oxygen consumption during the subsequential cycling performance.

Methods

Experimental Approach to the Problem

A parallel, randomized, controlled trial was conducted following the Consolidated Standards of Reporting Trials (CONSORT) 2010 recommendations (22,27) (Figure 1). Using an appropriate randomization test procedures, as the Excel-Based Statistical Package, that provides an analysis of level, trend, and variability in outcome assessments and effect-size estimates (24), subjects were randomly allocated to 3 groups: a 6-beat-kick group (K6), a 4-beat-kick group (K4), and a control group (CG). During the 8-week training program, using a pyramidal block periodization (with a progressive increase in volume and subsequently intensity) (32,36), subjects in the K6 group were asked to adopt 6 kicks for each swim stroke, those in the K4 group were asked to adopt 4 kicks for each swim stroke, whereas those in the CG were asked to maintain their regular kicking frequency. Before and after the training program, subjects were tested on a Mixed-Team Relay distance simulation, composed by a 400-m swim segment followed by a 10-minute cycle segment, with the concomitant measurement of biomechanical, physiological, and performance outcomes.

Subjects

Fifteen triathletes (13 men and 2 women; age range: 19–36 years) belonging to the same team were recruited to participate in this study (Table 1). They were sprint or middle-distance triathlon specialists with a minimum of 5 years of competitive experience at national level, and they were involved on a standard training schedule from 7 to 11 sessions per week. At the time of recruitment, none of them reported muscle and joint injuries during the past 6 months and none of them were taking drugs, medication, or dietary supplements, which could interfere with the experimental procedures. All subjects satisfied the criteria for

participating at national competitions with the medical certification demonstrating the absence of cardiovascular, pulmonary, metabolic, bone, or joint diseases. Subjects could not be blinded after allocation to one of the experimental groups; therefore, they were unaware of the goals of the study. They received instructions concerning the training program and any known risk, and they were also informed that they were free to withdraw from the experimental procedures at any stage. They gave their informed written consent. The study was approved by the Local Institutional Ethics Committee of Area Vasta Emilia Nord (AVEN, Emilia-Romagna Region, on April 13, 2021, protocol number: 1384/2020/SPER/UNIPR), and all experimental procedures were conducted in accordance with the Declaration of Helsinki (41).

Procedures

Experimental Training Protocol. A swimming training program was executed for 8 weeks, 2 times per week (with a range of 800–1,400 m for each training session) during the competitive period of the triathletes' annual training cycle. This program replaced the first part of their regular swimming training, which consisted of a total volume ranging from 3,000 to 3,500 m for each swimming training session. Moreover, the athletes maintained the same training program for cycling and running activities. An 8-week training program has been considered a minimal duration to induce training adaptations (14). During the training program, subjects were not involved in any official competition. The 3 groups performed a volume-equated training program, which differed only for the number of kicks for each swim stroke, with volume and intensity being progressively designed to increase the training workload. A detailed description of the training program is provided in Table 2.

Testing Session. A physiological and biomechanical profile was characterized by administering a Mixed-Team Relay distance test trial before (Pretraining) and after (Posttraining) the training program, consisted of a 400-m swim segment followed by a 10-minute cycle segment. Before the testing session, subjects were asked to refrain from any nutritional supplementation, to avoid exercise for 2 days before the session, and to abstain from alcohol and caffeine consumption during the 12 hours before the session. Pretraining and Posttraining testing sessions were performed at the same time of the day ($9 \text{ AM} \pm 30 \text{ minutes}$) and with similar environmental conditions (temperature: $24 \pm 3^\circ \text{C}$, humidity: $44 \pm 10 \text{ g}\cdot\text{m}^{-3}$). The swim segment was performed in a 25-m outdoor swimming pool (depth: 1.90 m, water temperature: $25 \pm 2^\circ \text{C}$), in the first lane and without any further swimmers in the other lanes, using a front crawl stroke technique and wearing a trisuit. The 10-minute cycle segment was performed on a cycle ergometer (Elite Directo-XR, Padua, Italy) with each subject's own bike. It was divided into a 3-minute all-out section, followed by a 7-minute race pacing effort with the attempt to replicate the critical power test. The explanation reflects the intermittent nature of the Mixed-Team Relay performance and the need for both aerobic and anaerobic capacities across various moments of the segments. Considering that the lactate threshold assessment might be less informative for this performance, the cycling Critical Power could better explain athletes' capability, which is generally evaluated with a 3-minute all-out test. During this form of test, the obtained parameters can have an implication for the assessment of physical fitness, the prescription of exercise training, and the prediction of performance during high-intensity exercise. Moreover, the test provides the

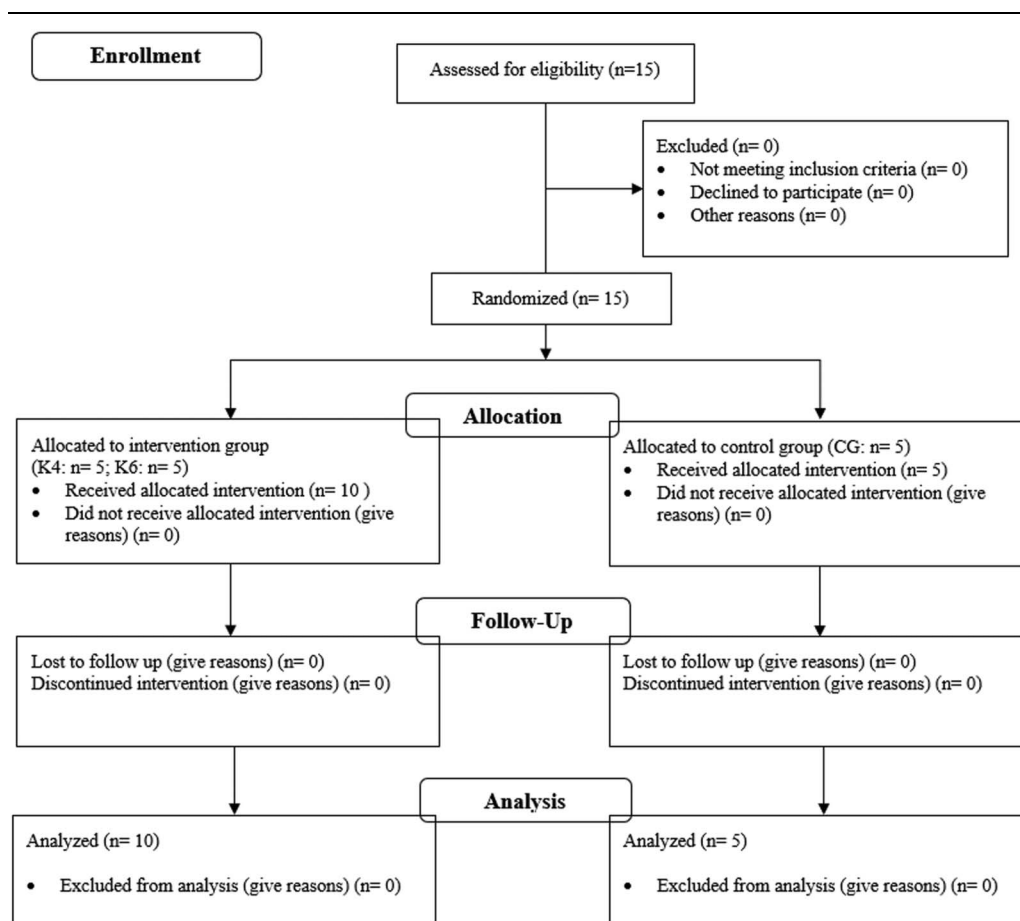


Figure 1. CONSORT 2010 flow diagram (27).

point where the $\dot{V}O_2$ begins to rise with a concomitant reduction in cycling power. The development of the $\dot{V}O_2$ slow component during a high-intensity constant work rate can lead to exhaustion approximately 45 seconds from the start of the test (33). Furthermore, because the slow component of $\dot{V}O_2$ and the fatigue process are related, evaluating the neuromuscular contribution to exercise during time trials could lead to a new interpretation on the influence of the swim segment on the cycle segment. Consequently, the above-mentioned explanation justifies the choice of the 3-minute all-out section and the 45-second portion of cycle segment.

Therefore, a concomitant evaluation of biomechanical and physiological parameters was conducted during the entire test trial. However, only for the last 100-m swim segment and for the first 45-second cycle segment, the data were extrapolated and processed for the statistical analysis.

Biomechanical Measurements. The investigation of biomechanical parameters during the swim segment was conducted

using 2 sport watch trackers (Garmin Forerunner 735XT, Olathe, KA) placed at wrist and ankle location. The Garmin Forerunner 735XT tracker integrates a triaxial accelerometer and magnetometer with a sampling frequency of 25 and 1 Hz, respectively, and has already been validated as a wearable kick-count device for triathletes (5). Moreover, the entire test trial was recorded using an underwater camera (GoPro HERO7+, San Mateo, CA) held parallel to the direction of swimming to further ascertain biomechanical parameters.

From the wrist tracker, stroke frequency (representing the total number of swim strokes), stroke length (representing the distance per swim stroke and calculated as the number of stroke frequency divided by 25-m lap), and average swim velocity (calculated as 25-m lap divided by 25-m lap time) were obtained for each 25-m lap of the last 100-m swim segment, and the average values were used for the statistical analysis. From the ankle tracker, kick frequency (representing the number of swim kicks) and kick length (representing the distance per swim kick and calculated as the

Table 1

Demographic characteristics of subjects at baseline level for the 3 groups (mean $\pm$ SD).*

Variables	K6 (n = 4 males and 1 female)	K4 (n = 4 males and 1 female)	CG (n = 5 males)
Age (y)	23.2 $\pm$ 3.1	27.4 $\pm$ 5.1	29.6 $\pm$ 3.5
Height (m)	1.7 $\pm$ 0.0	1.7 $\pm$ 0.1	1.8 $\pm$ 0.1
Body mass (kg)	62.3 $\pm$ 6.0	70.0 $\pm$ 8.3	73.4 $\pm$ 8.4
BMI (kg·m ⁻²)	20.8 $\pm$ 0.9	23.0 $\pm$ 1.5	22.4 $\pm$ 1.2

*BMI = body mass index; K6 = 6-beat-kick group; K4 = 4-beat-kick group; CG = control group.

Table 2**Detailed description of the 8-wk training program.*****Eight-week × 16 workout protocol (2 training per week)**

Frequency of swim kicks as per assigned group (CG/K6/K4) (front crawl)	200 m							
	4 reps × 50 m sub-maximal recovery 15 s							
First week	8 reps × 25 m (only swim kicks) recovery 10 s (number of swim kicks as per assigned group)							
	Second week	Third week	Fourth week	Fifth week	Sixth week	Seventh week	Eighth week	
First training (range of 200–800 m)	2 set × (4 reps × 25 m recovery 10 s) recovery 30 s between set	8 reps × 25 m recovery 5 s	4 reps × 50 m recovery 10 s	6 reps × 50 m recovery 5 s	4 reps × 100 m recovery 10 s	4 reps × 100 m recovery 5 s	2 set × (4 reps × 100 m recovery 5 s) and recovery 15 s between set	3 reps × 200 m recovery 5 s
Second training (range of 200–400 m)	2 set × (6 reps × 25 m recovery 10 s) recovery 30 s between set	10 reps × 25 m recovery 5 s	6 reps × 50 m recovery 10 s	2 reps × 100 m recovery 10 s	2 set × (2 reps × 100 m recovery 5 s) recovery 15 s between set	2 reps × 200 m recovery 10 s	1 rep × 300 m	1 rep × 400 m

*CG = control group; K6 = 6-beat-kick group; K4 = 4-beat-kick group.

number of kick frequency divided by 25-m lap) were obtained for each 25-m lap of the last 100-m swim segment, and the average values were used for the statistical analysis. Furthermore, the stroke index and kick index were calculated by multiplying the swim velocity by stroke length or kick length (11), respectively, for each 25-m lap of the last 100-m swim segment, and the average values were used for the statistical analysis.

During the cycle segment, power output and cadence were recorded using a cycle ergometer (Elite Directo-XR, Padua, Italy) at a sampling of 1 Hz. Maximum and average power output and minimum, maximum, and average cadence of the first 45-second cycle segment were used for the statistical analysis.

Physiological Measurements. Respiratory gas exchanges during the swim and cycle segments were measured breath-by-breath using a portable telemetric gas analyzer system (Cosmed K5, Cosmed, Rome, Italy). In particular, during the swim segment, the gas analyzer was attached to a swimming snorkel (Cosmed AquaTrainer, Rome, Italy). Breath-by-breath measurements were sampled to each second for the following parameters: O_2 uptake ($\dot{V}\text{O}_2$) kinetics, minute ventilation, and energy expenditure. According to standardized procedures (19) and manufacturer instructions, the Cosmed K5 system was calibrated before the swim segment after a 30-minute warm up period. In addition, HR was continuously monitored using a chest strap monitor (Garmin HRM-Tri 735XT, Olathe, KA), and data were recorded by the sport watch (Garmin Forerunner 735XT, Olathe, KA) at a sampling of 1 Hz. Maximum and average values for each parameter of the last 100-m swim segment and the first 45-s cycle segment were used for the statistical analysis.

Fingertip capillary blood lactate concentrations ([La]) were measured with Lactate Pro 2 (Arkaj, Tokyo, Japan) before and after the swim segment and at 3rd and 10th minute for the cycle segment.

Statistical Analyses

Data were analyzed using the Statistical Package for the Social Sciences, version 28.0 (SPSS Inc., Chicago, IL). The level of statistical significance was set at $p < 0.05$ for all computations, and all results are presented as mean $\pm$ SD. The normality assumption for each variable was verified using the Shapiro-Wilk test, which confirmed the normal distribution of data for the investigated variables. Consequently, a mixed analysis of variance (ANOVA) was applied to evaluate the main and interaction effects of the between-subjects factor group (K6, K4, CG) and the within-subjects factor time (Pretraining, Posttraining) on all the dependent variables. Effects sizes for main effects were calculated as partial eta squared (η^2) and interpreted as small (0.01 – 0.06), medium ($0.06 < \eta^2 < 0.14$), and large effects (> 0.14). In case of significant interactions, post hoc analysis using 1-way ANOVA and paired t test with Bonferroni's correction for multiple comparisons were applied. Cohen's d was calculated for the resulting effects from post hoc analysis and interpreted as trivial (< 0.19), small (0.20 – 0.59), moderate (0.60 – 1.19), large (1.20 – 1.99), very large (2.0 – 4.0), and extremely large effects (> 4.0) (10,13). The analysis of associations between the swim and cycle segments for biomechanical and physiological variables was determined by Pearson's product-moment correlations. The strength of associations was quantified and interpreted according to the following categories: ≤ 0.1 (trivial), 0.11 – 0.30 (small), 0.31 – 0.50 (moderate), 0.51 – 0.70 (large), 0.71 – 0.90 (very large), and ≥ 0.9 (nearly perfect).

Results

Descriptive Statistics

The performance outcomes and blood lactate concentrations during the swim and cycle segments for the 3 groups are presented in Tables 3 and 4. No differences emerged for the investigated variables.

Table 3**Performance outcomes during the Mixed-Relay Team test trial at Pretraining and Posttraining for each group (mean $\pm$ SD).***

Group	Swim segment completion time (s)		Cycle segment distance covered (km)	
	Pretraining	Posttraining	Pretraining	Posttraining
CG	321 $\pm$ 17.3	339.4 $\pm$ 40.5	7.41 $\pm$ 0.19	7.34 $\pm$ 0.22
K4	349.6 $\pm$ 30.0	345.4 $\pm$ 30.6	6.98 $\pm$ 0.43	7.01 $\pm$ 0.46
K6	363.6 $\pm$ 72.9	365.4 $\pm$ 79.8	6.81 $\pm$ 0.42	6.84 $\pm$ 0.50

*CG = control group; K6 = 6-beat-kick group; K4 = 4-beat-kick group.

Table 4**Blood lactate (mmol·L⁻¹) concentrations during the Mixed-Relay Team test trial at Pretraining and Posttraining for each group (mean ± SD).***

Group	Swim segment				Cycle segment			
	Pretraining		Posttraining		Pretraining		Posttraining	
	Before	After	Before	After	3rd min	10th min	3rd min	10th min
CG	1.86 ± 0.26	7.78 ± 3.61	2.26 ± 0.71	6.68 ± 2.77	14.84 ± 3.53	12.46 ± 2.94	10.92 ± 2.89	10.12 ± 4.61
K4	1.86 ± 0.25	6.78 ± 1.86	2.14 ± 0.46	7.24 ± 2.17	12.28 ± 3.69	11.66 ± 3.50	13.56 ± 4.53	13.82 ± 3.79
K6	2.04 ± 0.42	7.24 ± 3.86	1.78 ± 0.40	7.24 ± 2.89	11.66 ± 3.49	12.52 ± 4.60	11.14 ± 3.29	10.82 ± 3.53

*CG = control group; K6 = 6-beat-kick group; K4 = 4-beat-kick group.

Swim Segment

A time by group interaction emerged for stroke frequency, stroke length, stroke index, average energy expenditure, and minute ventilation (Table 5). Post hoc analysis with Bonferroni's correction ($p < 0.0083$) confirmed a lower stroke frequency ($p = 0.004$, effect size [ES] = 1.91) and a higher stroke length ($p =$

0.002, ES = 2.00) in K6 compared with CG at Posttraining. At Posttraining, nominal differences also emerged in K6 compared with K4 group for both stroke frequency ($p = 0.029$, ES = 2.57) and stroke length ($p = 0.017$, ES = 1.41), in K6 compared with K4 ($p = 0.024$, ES = 1.80) and control ($p = 0.025$, ES = 1.47) groups for average energy expenditure. For stroke index and

Table 5**Biomechanical and physiological changes during the last 100 m of the swim segment at Pretraining and Posttraining (mean ± SD).*,†**

Parameter	Group	Pretraining	Posttraining	Time effect p (ES)	Group effect p (ES)	Time × Group p (ES)
Stroke frequency (n)	CG	41.8 ± 5.5	49.6 ± 2.5	0.021 (0.371)	0.736 (0.050)	0.015 (0.503)
	K4	43.6 ± 5.2	48.4 ± 0.6			
	K6	48.4 ± 4.6	45.8 ± 1.3			
Stroke length (m)	CG	2.4 ± 0.3	2.0 ± 0.1	0.015 (0.401)	0.621 (0.76)	0.022 (0.472)
	K4	2.3 ± 0.3	2.1 ± 0.0			
	K6	2.1 ± 0.2	2.2 ± 0.1			
Stroke index (m ² ·s ⁻¹)	CG	3.0 ± 0.4	2.4 ± 0.4	0.007 (0.462)	0.336 (0.166)	0.014 (0.508)
	K4	2.7 ± 0.5	2.4 ± 0.2			
	K6	2.3 ± 0.5	2.4 ± 0.3			
Kick frequency (n)	CG	93.4 ± 28.9	101.2 ± 24.2	0.528 (0.034)	<0.001 (0.694)	0.478 (0.116)
	K4	92.2 ± 19.3	83.0 ± 23.3			
	K6	154.4 ± 17.1	142.0 ± 28.7			
Kick length (m)	CG	1.2 ± 0.4	1.0 ± 0.2	0.651 (0.018)	0.006 (0.579)	0.508 (0.107)
	K4	1.1 ± 0.2	1.3 ± 0.5			
	K6	0.6 ± 0.1	0.7 ± 0.2			
Kick index (m ² ·s ⁻¹)	CG	1.5 ± 0.5	1.2 ± 0.4	0.795 (0.006)	0.003 (0.630)	0.505 (0.108)
	K4	1.3 ± 0.3	1.5 ± 0.6			
	K6	0.7 ± 0.2	0.8 ± 0.2			
Average $\dot{V}O_2$ (ml·kg ⁻¹ ·min ⁻¹)	CG	40.7 ± 15.3	42.2 ± 12.2	0.493 (0.040)	0.904 (0.017)	0.232 (0.216)
	K4	35.4 ± 9.7	46.1 ± 5.1			
	K6	45.4 ± 7.4	40.6 ± 10.7			
Maximum $\dot{V}O_2$ (ml·kg ⁻¹ ·min ⁻¹)	CG	58.6 ± 13.4	54.7 ± 9.9	0.996 (0.000)	0.688 (0.060)	0.380 (0.149)
	K4	52.7 ± 9.7	60.8 ± 6.5			
	K6	62.7 ± 12.1	58.6 ± 13.9			
Average minute ventilation (L·min ⁻¹)	CG	79.3 ± 31.0	76.3 ± 24.0	0.078 (0.236)	0.783 (0.040)	0.065 (0.366)
	K4	56.9 ± 24.7	82.7 ± 14.2			
	K6	74.3 ± 15.5	78.2 ± 7.3			
Maximum minute ventilation (L·min ⁻¹)	CG	104.6 ± 27.5	91.0 ± 23.2	0.545 (0.031)	0.828 (0.031)	0.038 (0.420)
	K4	81.8 ± 17.5	98.5 ± 16.0			
	K6	91.4 ± 24.1	96.2 ± 15.4			
Average energy expenditure (kcal·min ⁻¹)	CG	14.6 ± 4.7	15.0 ± 3.0	0.909 (0.001)	0.364 (0.155)	0.042 (0.411)
	K4	11.9 ± 2.8	15.0 ± 1.9			
	K6	14.0 ± 2.6	10.8 ± 2.7			
Maximum energy expenditure (kcal·min ⁻¹)	CG	19.7 ± 3.6	19.8 ± 6.1	0.631 (0.020)	0.306 (0.179)	0.073 (0.354)
	K4	16.3 ± 3.0	18.6 ± 2.6			
	K6	17.8 ± 4.6	14.1 ± 3.9			
Average HR (b·min ⁻¹)	CG	151.4 ± 16.3	147.0 ± 13.4	0.307 (0.087)	0.132 (0.287)	0.303 (0.180)
	K4	161.1 ± 6.6	163.0 ± 9.0			
	K6	164.2 ± 10.5	161.5 ± 10.5			
Maximum HR (b·min ⁻¹)	CG	155.8 ± 16.1	150.6 ± 14.6	0.182 (0.143)	0.170 (0.256)	0.203 (0.233)
	K4	163.8 ± 6.3	165.2 ± 9.7			
	K6	167.6 ± 9.5	165.4 ± 9.8			

*HR = heart rate; CG = control group; K6 = 6-beat-kick group; K4 = 4-beat-kick group; ES = effect size calculated as partial eta squared.

†Bold values highlight a significant effect at $p < 0.05$.

Table 6**Biomechanical and physiological changes during the first 45 s of the cycle segment at Pretraining and Posttraining (mean ± SD).*,†**

Parameter	Group	Pretraining	Posttraining	Time effect <i>p</i> (ES)	Group effect <i>p</i> (ES)	Time × Group <i>p</i> (ES)
Average power (W)	CG	392.6 ± 54.9	381.7 ± 43.7	0.195 (0.136)	0.208 (0.230)	0.768 (0.043)
	K4	334.3 ± 48.7	328.1 ± 53.6			
	K6	348.8 ± 56.6	325.4 ± 67.8			
Maximum power (W)	CG	530.0 ± 80.3	486.2 ± 61.3	0.089 (0.222)	0.214 (0.227)	0.851 (0.026)
	K4	438.0 ± 44.5	417.8 ± 79.4			
	K6	447.0 ± 93.3	416.8 ± 118.8			
Average cadence (rpm)	CG	107.7 ± 12.0	104.0 ± 6.1	0.110 (0.199)	0.562 (0.091)	0.962 (0.006)
	K4	102.4 ± 6.0	99.4 ± 5.2			
	K6	104.3 ± 11.6	99.9 ± 7.9			
Maximum cadence (rpm)	CG	135.6 ± 25.4	126.2 ± 16.1	0.042 (0.302)	0.110 (0.307)	0.440 (0.128)
	K4	126.0 ± 18.9	108.6 ± 6.4			
	K6	112.6 ± 14.5	109.4 ± 9.2			
Minimum cadence (rpm)	CG	88.2 ± 7.4	77.8 ± 5.0	0.392 (0.062)	0.238 (0.213)	0.015 (0.505)
	K4	91.4 ± 5.6	81.6 ± 14.4			
	K6	83.8 ± 7.7	96.0 ± 6.7			
Average $\dot{V}O_2$ (ml·kg ⁻¹ ·min ⁻¹)	CG	49.4 ± 8.9	43.4 ± 7.2	0.001 (0.604)	0.499 (0.109)	0.004 (0.605)
	K4	42.3 ± 2.7	42.9 ± 5.3			
	K6	52.6 ± 5.0	33.6 ± 7.6			
Maximum $\dot{V}O_2$ (ml·kg ⁻¹ ·min ⁻¹)	CG	61.2 ± 8.8	48.8 ± 11.1	0.020 (0.376)	0.669 (0.065)	0.303 (0.180)
	K4	51.8 ± 1.6	50.5 ± 7.8			
	K6	61.2 ± 5.5	47.9 ± 16.9			
Average energy expenditure (kcal·min ⁻¹)	CG	18.1 ± 3.0	13.8 ± 3.2	<0.001 (0.759)	0.166 (0.259)	0.004 (0.606)
	K4	14.3 ± 2.1	13.9 ± 2.9			
	K6	16.4 ± 2.2	9.1 ± 3.1			
Maximum energy expenditure (kcal·min ⁻¹)	CG	22.8 ± 2.0	16.9 ± 3.6	<0.001 (0.690)	0.080 (0.344)	0.048 (0.397)
	K4	17.9 ± 2.4	16.7 ± 2.2			
	K6	19.4 ± 2.8	12.3 ± 4.8			
Average HR (b·min ⁻¹)	CG	138.4 ± 10.2	136.3 ± 15.6	<0.001 (0.691)	0.384 (0.147)	0.007 (0.564)
	K4	151.4 ± 7.4	139.6 ± 12.6			
	K6	152.7 ± 8.6	125.0 ± 8.2			
Maximum HR (b·min ⁻¹)	CG	151.2 ± 9.8	144.0 ± 17.2	<0.001 (0.651)	0.454 (0.123)	0.157 (0.266)
	K4	161.8 ± 4.5	149.0 ± 11.4			
	K6	163.6 ± 8.3	141.4 ± 11.7			

*HR = heart rate; CG = control group; K6 = 6-beat-kick group; K4 = 4-beat-kick group; ES = effect size calculated as partial eta squared.

†Bold values highlight a significant effect at $p < 0.05$.

maximum minute ventilation, post hoc analysis did not confirm any effect at Posttraining.

Cycle Segment

A time by group interaction emerged for minimum cadence, average HR, average $\dot{V}O_2$, and average and maximum energy

expenditure (Table 6). Post hoc analysis with Bonferroni's correction ($p < 0.016$) confirmed a reduction in K6 group between Pretraining and Posttraining program for average HR ($p = 0.005$, ES = 3.30), $\dot{V}O_2$ ($p = 0.014$, ES = 2.95), and energy expenditure ($p = 0.008$, ES = 2.72). Nominal differences were found in K6 group between Pretraining and Posttraining program for minimum cadence ($p = 0.039$, ES = 1.69) and maximum energy

Table 7**Analysis of the associations for biomechanical parameters between swim and cycle segment for each group at Posttraining.*,†**

Swim segment	Cycle segment	Groups		
		CG	K4	K6
		<i>r</i> (<i>p</i>)	<i>r</i> (<i>p</i>)	<i>r</i> (<i>p</i>)
Kick length (m)	Average power (W)	-0.960 (0.009)	0.346 (0.569)	0.751 (0.144)
	Maximum power (W)	-0.900 (0.037)	0.514 (0.375)	0.620 (0.265)
	Average cadence (rpm)	-0.493 (0.399)	0.289 (0.638)	0.954 (0.012)
	Minimum cadence (rpm)	0.066 (0.916)	0.447 (0.450)	0.935 (0.020)
	Maximum cadence (rpm)	-0.388 (0.518)	0.103 (0.870)	0.808 (0.098)
Kick index (m ² ·s ⁻¹)	Average power (W)	-0.943 (0.016)	0.404 (0.500)	0.805 (0.101)
	Maximum power (W)	-0.981 (0.003)	0.605 (0.279)	0.728 (0.163)
	Average cadence (rpm)	-0.430 (0.470)	0.267 (0.664)	0.954 (0.012)
	Minimum cadence (rpm)	-0.171 (0.784)	0.466 (0.429)	0.937 (0.019)
	Maximum cadence (rpm)	-0.124 (0.842)	0.161 (0.796)	0.888 (0.044)

*CG = control group; K6 = 6-beat-kick group; K4 = 4-beat-kick group.

†Bold values highlight a significant effect at $p < 0.05$.

expenditure ($p = 0.037$, $ES = 1.81$). Moreover, nominal differences emerged at Posttraining in K6 compared with K4 group for minimum cadence ($p = 0.036$, $ES = 1.28$) and average energy expenditure ($p = 0.030$, $ES = 1.60$) and in K6 compared with CG for average $\dot{V}O_2$ ($p = 0.042$, $ES = 1.32$), average energy expenditure ($p = 0.035$, $ES = 1.49$), and minimum cadence ($p = 0.011$, $ES = 3.08$).

Associations Between Swim and Cycle Segments

The analysis of the associations between swimming (last 100 m) and cycling (first 45 seconds) performance for biomechanical variables in each group after the training program is presented in Table 7.

Concerning the correlation for biomechanical parameters, significant correlations were obtained in the CG and the K6 group. For the CG, negative and almost perfect correlations emerged for the kick length and the kick index of the swim segment with respect to the average and maximum power values expressed during cycle segment. For the K6 group, positive and near-perfect correlations emerged for the kick length and the kick index of the swim segment with respect to the minimum, maximum, and average values of the cycling cadence during the cycle segment. No significant correlations emerged for the K4 group.

Discussion

The aim of this study was to evaluate the effect of an 8-week swimming training program on biomechanical and physiological parameters during the swim-to-cycle transition of a Mixed-Team Relay distance simulation. The results of this study showed that the use of a 6-beat-kick training program in high-level triathletes could improve swim stroke parameters, such as frequency, length, and index of efficiency, as energy expenditure. The changes in the swim stroke could be further linked to the improvements obtained in the cycling parameters, such as cadence, HR, oxygen consumption, and energy expenditure, which were found in the high-kick-frequency group (K6) compared with the other 2 groups (K4 and CG). The possible explanation of the influence of a prior performance (i.e., swimming) on a subsequent performance (i.e., cycling) could rely on the potentiation effect that a previous high-intensity exercise can have on a subsequent performance, and specifically when this is preceded by an exercise with the legs compared with an exercise with the arms (6,17). Indeed, the decreases in biomechanical (i.e., minimum cadence) and physiological (i.e., average $\dot{V}O_2$, average HR, average, and maximum energy expenditure) values observed during cycle segment in the K6 group are similar to previous findings that showed a high-exercise efficiency following a previous high-intensity exercise (7,8).

The evaluation of biomechanical parameters at Posttraining during swim segment demonstrated a significant increase in stroke frequency and a decrease in stroke length and stroke index in the CG. These results are in line with previous evidence that demonstrated an association of a shorter stroke length and a higher stroke frequency with a lower ability to generate the propulsion to overcome water resistance (12). Indeed, athletes with a poor swimming technique showed a 6.5% reduction in stroke length compared with athletes with a better technique (15). Therefore, swimming performance can be increased with the improvement of these swimming parameters, with longer training periods for stroke length, despite promoting energy saving (2),

and shorter training period for stroke frequency (28). This study found a reduction in stroke frequency and an increase in stroke length in the K6 group, demonstrating that an 8-week training program could be effective in high-level triathletes. Indeed, previous studies proved that the K6 group leads to an enhancement of the swimming technique because of the improvement in the coordination of the upper limbs and a better horizontal position in the water (and therefore less water resistance) (9). This improvement allows the maintenance of body stability, the enhancement of fluctuation in the rolling motion (38), and the increase in depth and length of swim stroke (34). Moreover, even if as only nominal differences, the results on swim kick variables between the K6 and CG groups could further confirm the improvement in the use of the kick. The changes observed in the CG demonstrated a worsening in the stroke performance. Based on the obtained results, it could be speculated that the lack of training adaptations in the CG might reflect the performance during the Mixed-Team Relay distance simulation, with a lower propulsive action. In fact, a higher kick frequency and a lower kick length might be also translated in a great effort as highlighted by a higher energy expenditure for the CG. Previous evidence already supported the detrimental effect of fatigue on swimming efficacy as indicated by a lower stroke length and a higher stroke frequency (23).

The evaluation of physiological parameters at Posttraining during swim segment demonstrated a greater reduction in the K6 group for energy expenditure, which could be linked to a better swimming economy because of a more efficient coordination of the limbs. Indeed, there is a direct relationship between the increase in stroke frequency, the decrease in stroke length, and the increase in energy costs during swimming (23,42).

The evaluation of biomechanical parameters at Posttraining during cycle segment highlighted an increase in the minimum cadence for K6 group. It is worth to mention that this result has been marked as a nominal difference because it did not reach the Bonferroni's correction. Although this study was not able to prove strong evidence, the potential impact of the training program on the parameter minimum cadence could be considered an initial and promising result. The relevance of this result could be attributed to the concept of the *optimal energy cadence*, a value explaining the optimization of cycling economy (40). Previous studies showed an *optimal energy cadence* around 60 rpm and a *freely cycling cadence* around 90 rpm (29,37). However, more recent studies recognized that *optimal energy cadence* and *freely cycling cadence* often overlap during high-intensity performance (39) and that maximal energy efficiency, with the optimal muscle recruitment and lower $\dot{V}O_2$, is between 80 and 100 rpm (25). This study demonstrated similar values for minimum cadence at Posttraining for the K6 group (96.0 ± 6.7 rpm). Therefore, it could be speculated that the high-kick-frequency training program could have induced adaptations on the K6 group. Those adaptations could be translated on a better capability to maintain an optimal cadence during the cycle segment even if under a condition of fatigue, when a higher kick frequency is maintained during the last portion of the prior swim segment. Furthermore, the improvement in the minimum cadence was accompanied by a significant decrease between Pretraining and Posttraining for the average HR, $\dot{V}O_2$, and energy expenditure and that the significant difference found between K6 and CG in the minimum cadence was accompanied by the nominal differences in the average $\dot{V}O_2$ and energy expenditure parameters. Therefore, the multidisciplinary nature of the triathlon performance requires a continuous adaptation in cycling cadence based on the various

phases of the competition, the position acquired in the previous swim segment (1), and in view of the final run segment (40).

The analysis of the associations showed a positive correlation between swim kick index and cycling cadence (for minimum, average, and maximum values) in the K6 group. Instead, in the CG, a negative correlation emerged between swim kick index and cycling power (for average and maximum values). It could be speculated that in the K6 group, an increase in swim kick efficiency might subsequently lead to an increase in the cycling cadence, reaching the values close to the *optimal energy cadence*, probably because of a greater muscle recruitment. Conversely, in the CG, the increase in swim kick might induce a decrease in power output in the subsequent cycle segment, which is hypothesized to be the result of a greater effort of athletes to produce optimal swim kick.

This study has some limitations that need to be addressed and can serve as guidance for future research. First, the small sample size might have influenced the magnitude of the effects because only 5 athletes were included in each group. However, small size is based on the need to recruit high-level triathletes, belonging to the same team and competing for sprint or middle-distance triathlon. Moreover, the nature of the study design, with the implementation of a specific training program, elevated the challenge to recruit different teams, thus limiting the sample size. Second, the investigation of gender effect was not possible. Third, it was not possible to extend the training program because of interference with the schedule of competition. Fourth, dietary intake of athletes was not controlled during the experimental procedures. However, we advised athletes to avoid changes in their eating habits during the entire training program.

In conclusion, the implementation of a specific training program centered on a higher kick frequency during the swimming performance seems to induce improvements in the swim stroke technique and enhance swimming economy and, subsequently, to allow the maintenance of a higher minimum cadence during cycle segment. This could be translated in a greater cycling economy, which could further influence the remaining triathlon performance.

PRACTICAL APPLICATIONS

The findings of this first study on the biomechanical and physiological changes during the swim-to-cycle transition could be translated in practical implications for training with the primary goal to induce improvement in swim efficiency and, subsequently, cycling economy. The improvement could be primarily attributed to the increase in swim stroke efficacy, as a consequence of the increase in the swim stroke length and the decrease in the frequency, with a lower energy cost during the cycle segment. Consequently, 8 weeks of a swimming training program centered on a higher kick frequency could be considered an effective strategy to enhance the swim-to-cycle transition. Therefore, triathlon coaches and athletes should consider incorporating this form of training within their regular training program, with a frequency of 2 times per week, with 400 m of high swim kick frequency (6 kicks for each stroke cycle), to enhance the swimming economy and consequently the cycling performance. The linear progression of the swim kick stimulation proposed by the current training program should lead to a greater and better use of swim kick and to an enhancement of the overall swimming technique. Moreover, coaches and athletes should include in their weekly training program also the simulation of

the swim-to-cycle transition, adopting the Mixed-Team Relay distance and using a high swim kick frequency (6 kicks for each stroke cycle) in the final part of the swim segment (last 100 m) and a high cycling cadence (from 90 to 100 rpm) in the initial part of the cycle segment (first 45 seconds). The enhancement of the swim-to-cycle transition could lead to an improvement in the work economy (swimming and cycling) and in the overall triathlon performance.

Acknowledgments

This work was supported by FIL-Fondi Locali per la Ricerca to GG. Grant Number GOBBI_G_FIL. The authors thank the Triathlon Coaches for the assistance with data collection. The authors acknowledge and thank the athletes for giving their time to complete this study. The results of this study do not constitute endorsement of the product by the authors or the National Strength and Conditioning Association. The authors have no conflicts of interest and no professional relationships with companies that will benefit from the results of this study. *Authors contribution:* L. Ambrosini and V. Presta research concept and study design and writing of the manuscript; E. Menegatti, A. Guarnieri and V. Bianchi literature review; V. Bianchi, A. Guarnieri, and G. Condello data collection; G. Gobbi and M. Vitale data analysis and interpretation; G. Gobbi, I. De Munari and G. Condello statistical analyses; E. Menegatti, M. Vitale and I. De Munari reviewing/editing a draft of the manuscript.

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